LESSON PLAN

AIRCRAFT WIRING PRACTICES (ELECTRICAL WIRING AND INTERFACE SYSTEM) (12)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Control horizontal and vertical movement, control of RPM, torque reaction, tail rotor compensations, contra rotating rotors.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Control horizontal and vertical movement, control of RPM, torque reaction, tail rotor compensations, contra rotating rotors.
5. Specific Objectives	12.1.4.1 Trainee will be able to explain horizontal and vertical movement, torque reaction, tail rotor compensations, contra rotating rotors.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of cyclic and collective control inputs, torque reaction principles, tail rotor compensation mechanisms, and contra-rotating rotor configurations. Safety: Ensure no loose items near workspace. Instructor supervision: Monitor trainee understanding of rotational dynamics.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary functions of the cyclic and collective controls?
(b) How does torque reaction affect the helicopter fuselage?
(c) Why are contra-rotating rotors used in certain helicopter designs?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Horizontal and vertical movement control • Control of RPM via collective pitch • Torque reaction and fuselage rotation	L & Ds	PPT, IB	15 min
(b)	• Tail rotor compensations for torque • Pedal control for yaw movement • Efficiency of tail rotor systems	L & Ds	PPT, PDF	20 min
(c)	• Contra rotating rotors design • Cancellation of torque effects • Advantages of coaxial systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the tail rotor compensates for torque reaction.
(b) Describe the mechanism of contra-rotating rotors in eliminating torque.
(c) How does the pilot control vertical movement using the collective?

12. Home Assignment and Follow Up	Prepare a brief note on the differences between single main rotor and contra-rotating rotor systems, referencing concepts covered in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Main rotor movements, symmetry of lift feathering, flapping, dragging, hook's joint effect, flapping to equality, phase lag and advance angle concepts.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min